THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 1: **Patterns in Mathematics** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The sum of the first 35 odd numbers is 1225. The sum of the odd numbers from 31 to 69 is: 1

(a) 1001 **(b)** 1000 **(c)** 999 **(d)** 1015

2. The difference between the squares of the 25th and 24th numbers is: 1

(a) 51 **(b)** 48 **(c)** 49 **(d)** 50

3. The first term of a pattern is 12 and the 17th term is 220. The common difference is: 1

(a) 14 **(b)** 15 **(c)** 13 **(d)** 12

4. A pattern adds 1, then 2, then 3, ... Starting at 25, the term after 15 additions is: 1

(a) 145 **(b)** 144 **(c)** 160 **(d)** 146

5. The sum $1^3 + 2^3 + \cdots + 10^3$ equals: 1

(a) 3025 **(b)** 3024 **(c)** 3080 **(d)** 3026

6. The 57th triangular number minus the 56th triangular number is: 1

(a) 57 **(b)** 58 **(c)** 56 **(d)** 59

7. In a party 20 people greet each other once with a hug. Number of hugs: 1

(a) 170 **(b)** 210 **(c)** 190 **(d)** 191

8. The sum $1 + 2 + 3 + \cdots + n$ equals 1378. Then n is: 1

(a) 53 **(b)** 51 **(c)** 54 **(d)** 52

9. The sum of the first 40 odd numbers is 1600. The sum of the odd numbers from 25 to 79 is: 1

(a) 1468 **(b)** 1456 **(c)** 1455 **(d)** 1457

10. The difference between the squares of the 30th and 29th numbers is: 1

(a) 59 **(b)** 58 **(c)** 61 **(d)** 60

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The sum of the first n odd numbers is 1444. Find n and the 38th odd number. 2

12. Find the sum of odd numbers from 23 to 55. 2

13. 17 cricket teams play a league in which every team plays every other team once. Find the number of matches. 2

14. The sum of the first n odd numbers is 1849. Find n and the 43th odd number. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In a pattern the 1st term is 6, the 2nd is 19, and so on by adding 13. Which term of the pattern equals 409? Find also the term just before it. 3

16. Sums of consecutive odd numbers give squares. Find the sum of the odd numbers from 1 to 59 and use it to find 30^2. Then evaluate the sum from 61 to 75. 3

17. A tower is built with rows of cubes: row 1 has 1 cube, row 2 has 3, row 3 has 6, and so on (triangular numbers). How many cubes are in row 8, and in the first 8 rows together? 3

OR

In a pattern the 1st term is 31, the 2nd is 43, and so on by adding 12. Which term of the pattern equals 355? Find also the term just before it. 3

18. Sums of consecutive odd numbers give squares. Find the sum of the odd numbers from 1 to 69 and use it to find 35^2. Then evaluate the sum from 71 to 83. 3

OR

A tower is built with rows of cubes: row 1 has 1 cube, row 2 has 3, row 3 has 6, and so on (triangular numbers). How many cubes are in row 9, and in the first 9 rows together? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Look at the pattern: the 10th figure has 43 dots. How many dots are in figure 18? If a figure has 95 dots, which figure is it? Show that the dots form a pattern with common difference 4. 5

OR

Two sequences: A is $8, 13, 18, 23, \ldots$ and B is $2, 11, 20, 29, \ldots$. Find the 11th term of each and decide which is larger and by how much. 5

20. Look at the pattern: the 5th figure has 47 dots. How many dots are in figure 15? If a figure has 212 dots, which figure is it? Show that the dots form a pattern with common difference 11. 5

OR

Two sequences: A is $11, 17, 23, 29, \ldots$ and B is $6, 14, 22, 30, \ldots$. Find the 14th term of each and decide which is larger and by how much. 5

END OF QUESTION PAPER — Total: 40 Marks

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